



*Advanced AI-based dynamic Integration
of heterogeneous data for semantic
clusters formalisation in human-centric
intelligent enterprise systems*

Creation of a logistic ontology & Dataset generation

Mario Lezoche

Thomas Pallet

Tuesday 24th March 2026

Table of content

01

Introduction

02

Ontology

03

Dataset

04

Conclusion

Introduction

IOF Core as the foundation of the ontology

<https://github.com/iofoundry/ontology/tree/master>

Usage of

- MaterialEntity
- Agent
- Process

All disjoint (mutally), process is not a physical object and agent not a location

Introduction

Built then classes for those 3 cores

MaterialEntity :

subclass

LogisticsItem (things moved/stored)

- Products
 - RawMaterial
 - FinishedProduct
 - SparePart

PhysicalLocation (things that contain/locate other things)

- Warehouse
 - StorageZone
 - Shelf
 - Bin
 - Dock

Agent :

- LogisticAgent
 - Worker
 - Robot

Process :

- LogisticsProcess
 - TransportProcess
 - StorageProcess
 - PickingProcess

RawMaterial, FinishedProduct, SparePart disjoint (mutually)

same for LogisticsItem, PhysicalLocation, LogisticAgent

Introduction

Defined six forward properties (moves, performedBy, storedIn, locatedIn, hasOrigin, hasDestination)
each with a declared inverse (isMovedBy, performs, hosts, contains, isOriginOf, isDestinationOf).

SKOS (Simple knowledge organization System)

Wanted to combined the OWL syntax I had for now and add SKOS

SKOS use for taxonomy, in my case, because I will discover family and subfamily characteristics when reading documentation, I need to add them.

To do so, using skos only for this part can be good I think, I read stuff about SKOS and I am not sure that it is a good idea but we will try it and if it is not good, then we will still use OWL but for the whole ontology and the taxonomy won't be as precise as what skos are but we will see

Table of content

01

Introduction

02

Ontology

03

Dataset

04

Conclusion

Ontology creation

Performed an interview with the client (Gauthier) in 3 steps :

Phase 1:

- Can you describe this file and what these columns represent?

Phase 2:

- For each column, can you explain concretely what it represents?
- Are there relationships between some columns?
- How do family and subfamily work (taxonomy, classification logic)?
- What units are used (dimensions, weight)?
- What does annual_sale represent?
- What is the relationship between annual_sale and ABC_consumption?

Ontology creation

Phase 3:

- Is Reference a unique identifier?
- Can there be duplicates?
- Can multiple products have the same designation?
- Is the designation sufficient to understand the product?
- What is the exact relationship between family and subfamily?
- Do the values (e.g., 51, 13) have a meaning or are they arbitrary?
- Which columns should be interpreted together?
- Are there strong dependencies between some columns?
- Quantity: is it fixed or variable?
- Weight: what unit?
- Length / width / height: what units?
- How is annual_sale calculated?
- How is ABC_consumption determined?
- What is the exact relationship between annual_sale and ABC_consumption?
- Is it a fixed or evolving rule?
- What relationships are “obvious” to you but not present in the data?
- What is missing to fully understand a product?

Interview

Done on Teams, raw txt files, split it in 3 phases, then correct it and translate it in english

Now, ontology is done, next step is the to make sure that the ontology is correct, when parsing like the txt files, make sure the correct relations are detected

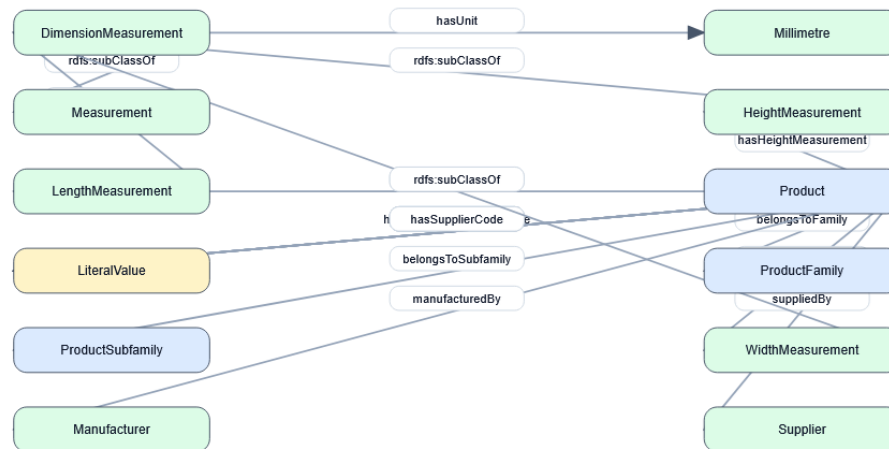
This NLP processing then is about having one general ontology - 2 levels of knowledge then by reading documentation, it makes it schema-level changes like (Dimension, hasUnit, Millimetre)

and parsing CSV, instance-level :
(Product_ABXDDF1044, hasLengthMm, 276.0)
BOS10441044

Interview results

I build then an automatic way with rule based patterns to detect relations automatically (so no validator it just raw extraction) just to make sure my ontology was stable and that it contained everything we needed for the task

Cumulative graph at phase1



Interview results

Problems :

Ambiguity in text like:

- Dimension should be in centimeter or millimeter, then what should we do about that ? My ontology can infer both of this, is this not okay ?
- Same with weight obviously, gram / kilogram
- Information such as that the reference of the article is composed of some first digit that gives us the supplier name and then the rest of it is the article id (but article id not unique because some are the same without the supplier name !)
- Problem of unit, it says quantity = 4 but in fact it depends on 'conditionnement', like it can be quantity of 4 individual product or it can be 4 element of pack of 10 individual product !
→ Big problem because it depends every time, so we can't say anything about that.

Table of content

01

Introduction

02

Ontology

03

Dataset

04

Conclusion

Next steps

Next step should be the dataset creation

Question about then doing a dataset which is actually a documentation ?
Because for the CSV, no need for NLP preprocessing.

Problem might be that I have a thin vision about the problem ?
(but I had like other client's so maybe check on those one and look for documentation for those one too ?)

What's the point of the CSV adaptation or

Table of content

01

Introduction

02

Ontology

03

Dataset

04

Conclusion

Thank you for your attention !

Questions ?